Describing the Dataset and Source Reputability

For Module One, I've decided to go with the "Leading Causes of Death: United States" dataset, sourced directly from the National Center for Health Statistics (NCHS). Upon evaluating the validation of these sources, we need to remember that data is the digital persistence of facts, knowledge, and information consolidated for reference or analysis (Wintjen, 2020). The NCHS is apparently a highly reputable source that utilizes the National Vital Statistics System to collect standardized death certificate filings from across the country (Centers for Disease Control and Prevention, 2025). This ensures that the data is a reliable collection of information for national reference. Jafari (2022) identifies the most universal data structure as the table, and datasets cohere to that format through a clear formation of rows and columns. Within this table, the data types include nominal values such as "State" and "Cause Name," which serve as qualitative labels; discrete numerical values for "Year" and "Deaths"; and continuous ratio data for the "Age-adjusted Death Rate," which allows for normalized comparisons across varying population sizes (Jafari, 2022).

Identifying Data Quality Issues

Several data quality issues were identified during the data screening process analysis step that could impact the reliability of the analysis. A main issue involves some missing or suppressed values in the "Age-adjusted Death Rate" column, which occurs when death counts in specific regions are too low to be statistically significant. As Wintjen (2020) notes, the reality of working with data is rarely straightforward, and there are "data gaps" that require the use of specific techniques to discover useful information without compromising the integrity of the findings. In addition, the "State" column contains a collection of "United States" entries mixed with individual state data, which is a structural consistency issue. Following Jafari's (2022) principle that data is "useless until used with regard to higher-level conventions and understandings," failing to identify these inconsistent formats and collection rows would lead to significant errors in calculation, as analysts must understand the statistical meaning of patterns being observed.

Overall Quality and Cleanliness

For the most part, the NCHS dataset is high in quality and cleanliness, though it's not perfectly "clean" if left alone. The data structure seems to be quite conformant to standard analysis requirements, providing the necessary "attributes" to answer the who, when, and what of mortality trends (Wintjen, 2020). Some specific examples of what is high quality include the standardized ICD-10 headers used for causes of death. Though cleanliness issues arise from the overlap between broad and specific cause categories, this requires some time to cleanse and transform. For instance, if an analyst isn't able to filter for specific hierarchies within the "113 Cause Name" column, they might double-count mortality figures. This demonstrates that high-quality data still requires data literacy, as Wintjen (2020) describes it: the ability to read, work with, analyze, and effectively argue with the data.

Impact of Data Quality Issues on Analysis

Identified data quality issues significantly impact the ability to analyze the data effectively if they are not addressed during the initial stages of the "data-to-action" process. The presence of national aggregates within state-level columns acts as "noise" that can skew statistical summaries and lead to inaccurate visualizations. In addition, the suppressed rates in smaller populations limit the ability to gain what the DIKW hierarchy described as "Wisdom" regarding rare health conditions in specific regions. As Wintjen (2020) notes, data analysis is a craft and a journey where the analyst must solve inherent challenges to provide accurate insights. Having a robust preprocessing strategy, as outlined by Jafari (2022), is therefore required to ensure that the resulting insights are both accurate and reproducible, moving data from a raw state into a form that can support reliable conclusions.

Questions and Insights for Real-World Application

Essentially, this dataset was designed to help answer vital questions regarding public health and national security. Analysts are given the opportunity to determine which causes of death are trending upward and which geographic regions are successfully mitigating specific high risks through policy or intervention. Going from raw death counts (data) to regional trends (information) and policy insights (knowledge), the dataset serves as a powerful tool for informed decision-making. Through the application of practical data and analysis techniques, we can transform these digital facts into meaningful outcomes that provide real-world value (Wintjen, 2020). These insights allow for the identification of benchmarks in public health and help determine where resources should be allocated to address emerging mortality trends.

References

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